

BobaLink Browser Extension “ Status

Revision: 8 **Last modified:** 2026-06-10T23:00:00Z **Scope:** BobaLink (extension/) “ WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs and forwards them to the Boba merge service on port 7187. **Authority:** master plan docs/browser_extension/IMPLEMENTATION_PLAN.md (9 phases).

Captured-evidence-driven (11.4.5 / 11.4.45). Every PASS cites a real commit hash and/or a verified test/file artifact. Rows lacking runtime evidence are marked IN-PROGRESS or PENDING “ never PASS (11.4.6 no-guessing).

Baseline facts (verified this session “ Session 11)

- **HEAD:** 2011810 (pushed); wave-4 (Phase-9 ci-ext.sh gate + per-store zips, ru locale, content-XSS, credential-leak audit, decrypt-throw parity, tab-group Challenge, and the _locales packaging fix) staged for the next commit.
- **Extension test corpus: 527 Vitest tests across 49 spec files “** same-session npx vitest run “ Tests 527 passed (527) “ under extension/tests/{unit,perf,stress,chaos,integration,security,e2e} + src/**/*.npx tsc --noEmit clean; npm run lint 0 errors / 0 warnings. (+126 tests over the prior 287/22 baseline: integration 7, security 4 files, stress/chaos 10, locale 4, token suites 8, tab-groups 13, ally 18, group-send nits 2, +more.)
- **WXT build wiring “ COMPLETE.** WXT entrypoints exist at extension/src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html} (thin wrappers over the existing logic modules, so all prior test imports stay valid). npx wxt build produces a **loadable** extension/.output/chrome-mv3/ AND firefox-mv2/:manifest_version:3, background.service_worker:background.js, popup.html, options.html, content-scripts/content.js, icons 16/32/48/128, AND _locales/{en,ru}/messages.json (REQUIRED because default_locale:"en" + __MSG_*__ “ see the 11.4.138 note below). **Every manifest-referenced asset + the default-locale catalog verified present on disk** by the ci-ext.sh manual gate (11.4.38). Content-script matches are derived from the curated SITE_SELECTORS (24 hosts, **no** <all_urls>); permissions least-privilege (no scripting/tabs/cookies; +tabGroups for Phase-5); host_permissions = http://localhost:7187/* only; CSP script-src 'self'. Per-store zips (boba-link-1.0.0-{chrome,firefox,sources}.zip) produced by ci-ext.sh.

Per-phase status

Phase	Title	Status	Evidence
1	Foundation & scaffolding (WXT config, TS, shared libs, types, constants)	PASS	Scaffolded @33a9815; shared-lib + constants + type-guard unit specs green @15a9a61
2	Core detection & parsing engine (parsers, scanners, infohash dedup)	PASS	Parsers @7225470/fa03323; SHA-1 infohash + link/text scanners + perf/stress @fa03323;

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			scanner orchestrator cross-scanner dedup @946c61e
3	Extension shell “content / background / popup / options	PASS	Shell @e8fde43; background service-worker message router capstone @15a9a61
â€”	WXT build wiring (entrypoints/, wxt build “ .output/, manifest validate)	PASS	Entrypoints src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html}; npx wxt build “loadable .output/chrome-mv3/ (8/8 manifest assets verified present, Â§11.4.38); matches derived from SITE_SELECTORS, no <all_urls> (Session 11 @5edf6ac, pushed)
4	Boba backend integration (real client, queue auth, BE-1/2/3)	IN-PROGRESS	API leaf @e8fde43. Phase-7 decrypt-before-send wired (Session 11): BobaClient.create() decrypts the encryptedBobaApiToken bundle via shared/crypto; background reads the session passphrase from chrome.storage.session and sends the decrypted plaintext (default-open when locked). Token suites boba-client-token.test.ts 5 + background-token.test.ts 3 green. PENDING: live-7187 integration (require_backend(7187)), end-to-end detect“send“torrent-in-qBittorrent on the real backend
			Wired (Session 11): src/tabgroups/index.ts (dedupe across a group + batch dispatch, 13 tests) integrated into background/index.ts MENU_SEND_GROUP (deduped group batch “one addMagnets POST); manifest +tabGroups (MINIMAL “research-confirmed tabs NOT needed since only tab.id is read; Â§11.4.120 security-test reconciliation, mutation-verified). New background.test.ts MENU_SEND_GROUP test (RED-verified: handler no-op “FAIL). Independent review: GO-with-nits . Group-send nits (a) offline-queue enqueue-on-failure, (b) network-error notification, (c) hardened async flush “all FIXED (Session 11, RED“GREEN + 3—deterministic, independent review GO). TRACKED: decrypt-throw (wrong passphrase) on group send not enqueued/notified “pre-existing parity with Send-All, future phase
5	Tab-group batch (chrome.tabGroups “per-tab scan “batch send)	IN-PROGRESS	i18n: locale.test.ts (en completeness) + ru+de+fr+es+it+pt+ja locales added (8 total “plan target reached, 29-key parity, packaged chrome+firefox) with generalized locale-
6	UI/UX, i18n, accessibility, themes	IN-PROGRESS	

Phase Title	Status	Evidence
7 Security & credentials (delegate-by-default, no embedded key, log redaction)	IN-PROGRESS	parity.test.ts (key + placeholder parity, mutation-proven); a11y: tests/a11y/{popup,options}.a11y.test.ts (18 tests â€” roles/accessible-names/tablistâ†”i,Žtabpanel/live-regions, mutation-proven). tablist keyboard-nav (WAI-ARIA Arrow/Home/End). PENDING: deeper WCAG (contrast/focus), theme-switch evidence crypto.ts adopted; decrypt-and-send path landed + decrypt-throw on wrong passphrase now enqueues+notifies (parity, REDâ†”GREEN); session passphrase from chrome.storage.session, no embedded key, plaintext/passphrase never logged; security suite tests/security/* (least-privilege manifest, CSP, no-hardcoded-secret, secret-storage, content-XSS â€” render path proven safe via innerHTML-mutationâ†”FAIL); Â§11.4.10.A credential-leak audit challenges/security/credential_leak_audit.sh (PASS, mutation-verified); message-router robustness (16 tests) + isValidScanResult scan-result trust-guard + resolveTargetTabId cross-tab confused-deputy guard (fixed a real content-script info-leak: forged payload.tabId could read another tabâ€™s detections; sender.tab.id now wins) + scanner-hostile-input (19) + sender-trust (10) + content-XSS. PENDING: full pen-test suite (sender-origin validation, rate-limit) 527 tests/49 files green (unit/perf/stress/chaos/integration/security/e2e). Challenge challenges/extension/detect_and_forward_challenge.sh drives the REAL orchestrator+client end-to-end, PASS on captured evidence, mutation-verified (no-op stub â†”FAIL). HelixQA boba-bobalink.yaml BOBA-LINK-007 (detectâ†”forward) + BOBA-LINK-008 (tab-group batch) added; symlinked into challenges/helixqa-banks/. tab-group + detectâ†”forward Challenges both mutation-verified. E2E tests/e2e/extension-loads.spec.ts is a real MV3-load test, operator-gated SKIP in this headless sandbox (extension load unsupported; Â§11.4.3). PENDING: the full 13-type coverage matrix + live-backend integration + the coverage ledger to 100%
8 Testing to 100% (all types) + Challenges + HelixQA	IN-PROGRESS	Manual gate extension/ci-ext.sh (Session 11, Â§11.4.18 doc'd): tsc â†”lint â†”full vitest â†”chrome+firefox builds â†” Â§11.4.38
9 Build, packaging & distribution (manual â€” NO CI/CD)	IN-PROGRESS	

Phase Title	Status	Evidence
		artifact-verify (opens the manifest, asserts every referenced asset + the default_locale catalog exist non-zero) â†’ per-store wxt zip. CI-EXT: PASS â€” produces loadable chrome-mv3/ + firefox-mv2/ + bobalink-1.0.0-{chrome,firefox,sources}.zip. PENDING: store-listing metadata/submission + Â§11.4.65 user/dev/install doc siblings. NO CI/CD (manual only)

Status legend

- **PASS** â€” implemented and backed by a cited commit and/or verified test/file evidence.
- **IN-PROGRESS** â€” partially landed; some sub-tasks done with evidence, remainder explicitly enumerated as PENDING.
- **PENDING** â€” not yet started / no runtime evidence; planned in IMPLEMENTATION_PLAN.md.

Anti-bluff notes (Â§11.4 / Â§11.4.6 / Â§11.4.69)

- The 527 figure IS a same-session recorded `npx vitest run` result (Tests 527 passed (527)), not merely a static `grep` â€” it supersedes the prior 287/22 `grep-count` baseline. `tsc + lint` captured clean in the same session.
- The Â§11.4.38 loadable-artifact claim is verified by `ci-ext.sh` **opening the produced** `.output/chrome-mv3/manifest.json` and confirming every referenced asset AND the `default_locale` catalog exist non-zero on disk â€” not by "build exited 0".
- **Â§11.4.138 bluff-audit (corrected this session).** An earlier Status revision claimed the artifact was loadable on an "8/8 referenced assets present" check. That check was INCOMPLETE: the manifest declares `default_locale: "en" + __MSG_*__`, but the build packaged `no_locales/` (WXT copies static assets from `src/public/`, while the catalog lived at `src/_locales/`) â€” so Chrome would have REJECTED the extension at load. Caught by the new `ci-ext.sh` Â§11.4.38 step (which checks the `default_locale` â†’ `_locales/` invariant). Root cause fixed by moving the catalog to `src/public/_locales/{en,ru}/`; the rebuilt artifact now packages it (verified on disk). Permanent guard = `ci-ext.sh` (the regression can't recur silently). This is the Â§11.4.108 lesson: `build-exit-0` â‰‰ deployable-artifact.
- The Phase-7 decrypt path is proven by REDâ†’GREEN: ciphertext-on-the-wire (pre-fix) â†’ decrypted-plaintext header (post-fix), with the ciphertext never sent and the token never logged (`background-token.test.ts`, `boba-client-token.test.ts`).
- The detectâ†’forward Challenge was independently re-run and its no-op-stub negation verified (stubbed orchestrator â†’ FAIL) during code review.
- Independent multi-lens subagent code-review (correctness/security/build/docs) was run; the security + correctness + build + docs lenses returned GO / GO-with-nits. (The correctness/security/build subagent lenses were re-run in the conductor context after a transient platform throttle on subagent dispatch; an independent subagent re-pass runs before any release tag per Â§11.4.40.)
- **Two real bugs caught by the new tests + fixed (wave-5, REDâ†’GREEN, independently re-verified):** (1) a `popup`â†’`i`, `background` CONTRACT bug the integration round-trip exposed â€” the `popup` read `r.torrent.id` but the real background returns a flat `SendOutcome {id, â€¦}`, so a SUCCESSFUL send threw in the `popup` and showed a FALSE "Send failed" (row never flipped to Sent); the `popup-unit`

fake had masked it with the wrong shape. Fixed (popup reads `r.id`; fake corrected; integration assertion reconciled §11.4.120). (2) a `scan-result` TRUST bug the message-router robustness tests exposed – a hostile content script could send `{items:<non-array>}` and overwrite a tab's good detection set with garbage; fixed with an `isValidScanResult` shape-guard (16 robustness tests; the non-array test seeds-good – corrupt and asserts the good set survives).

- de/fr locales added (4 total, all 29-key parity, packaged in both chrome+firefox builds); orchestrator + rate-limiter stress (evidence-captured, deterministic); popup/options – i, Ž background integration round-trips (real `sendMessage` – onMessage bridge); USER/INSTALL/DEV guides (§11.4.65 exports).